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A Centre of Excellence in Science and Technology by the Catholic Archdiocese of Trichur

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

MICRO PROJECT REPORT

EXAM HALL SMARTPHONE AND BLUETOOTH DETECTOR

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY (B.Tech)

in

COMPUTER SCIENCE & ENGINEERING (CYBER SECURITY)

of

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY

CREATING TECHNOLOGY LEADERS OF TOMORROW

OCTOBER 2026

DECLARATION

We hereby declare that the report on "USB Attack Detection and Auto Isolation System" , submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Ms.Sagna L T. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. We understand that any violation of the above will be a cause for disciplinary action by the Institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously used by anybody as a basis for the award of any degree, diploma or similar title of any other University.

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BONAFIDE CERTIFICATE

This is to certify that the seminar report entitled **Exam Hall Smartphone and Bluetooth Detector** submitted by **ADORN PRADEEP, AISWARYA P S, JOYAL GEORGE JOSEPH, PRATHISHA V R, VISMAYA P S** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of **Bachelor of Technology** degree in **Computer Science & Engineering(Cyber Security)** is a bonafide record of the Micro Project carried out by her under our supervision and guidance.

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1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
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10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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- **PEO 2 :** Graduates will pursue successful careers as professionals, innovators or entrepreneurs engaged in technology development, deployment or engineering system implementation in industry and society and contribute to the economic growth of the country.
- **PEO 3 :** Graduates will demonstrate ethical responsibility, leadership, teamwork, and effective communication while adapting to evolving global technological environments through lifelong learning.

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- **PSO 1 :** Apply cyber security principles to mitigate cyber risk and analyze security requirements of an organization.
- **PSO 2 :** Develop and deploy secure software and infrastructure using modern tools and industry standards.
- **PSO 3 :** Uphold ethical, legal, and societal responsibilities while protecting privacy, data, and digital assets.

Course Outcomes (COs)

- **CO1 :** Explain the architectural features and instructions of the ARM microcontrollers. (K2)
- **CO2 :** Develop applications involving interfacing of external devices and I/O with ARM microcontroller. (K3)
- **CO3 :** Usevarious communication protocols of interaction with peer devices and peripherals. (K3)
- **CO4 :** Demonstrate the use of a real time operating system in embedded system applications. (K3)
- **CO5 :** Apply hardware security features of ARM in real world applications. (K3)

PO - CO Mapping

co	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PSO4
PBCST504.1	3	2	1	1	3	-	-	-	-	-	-	2	2	-	-
PBCST504.2	3	2	3	2	3	1	-	-	1	1	1	2	3	1	-
PBCST504.3	3	2	2	2	3	1	-	-	2	1	1	2	3	1	-
PBCST504.4	3	2	3	2	3	1	-	-	2	1	1	2	3	1	-
PBCST504.5	3	3	3	2	3	2	1	2	1	1	2	3	3	2	-

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Last but not least I extend my gratefulness to all teaching and non teaching staffs who directly or indirectly involved in the successful completion of this seminar and to all my friends who have patiently extended all sorts of help for accomplishing this undertaking.

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

1. No Poverty – End poverty everywhere.

2. Zero Hunger – End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.

3. Good Health and Well-being – Ensure healthy lives and promote well-being.

4. Quality Education – Ensure inclusive and equitable education for all.

5. Gender Equality – Achieve gender equality and empower all women and girls.

6. Clean Water and Sanitation – Ensure access to water and sanitation for all.

7. Affordable and Clean Energy – Ensure access to sustainable energy.

8. Decent Work and Economic Growth – Promote sustainable economic growth and decent work.

9. Industry, Innovation, and Infrastructure – Build resilient infrastructure, foster innovation.

10. Reduced Inequalities – Reduce inequality within and among countries.

11. Sustainable Cities and Communities – Make cities safe, resilient, and sustainable.

12. Responsible Consumption and Production – Ensure sustainable consumption and production patterns.

13. Climate Action – Take urgent action to combat climate change.

14. Life Below Water – Conserve and sustainably use oceans and marine resources.

15. Life on Land – Protect, restore, and promote sustainable use of ecosystems.

16. Peace, Justice, and Strong Institutions – Promote peaceful and inclusive societies.

17. Partnerships for the Goals – Strengthen global partnerships to support implementation.

SDG ADDRESSED BY THE MICROPROJECT AND JUSTIFICATION

PROJECT TITLE	**USB Attack Detection and Auto Isolation System**
SDG 4	• **SDG 4 – Quality Education** Provides practical experience in cybersecurity and embedded systems. • Develops skills in microcontrollers and hardware security. • Encourages innovation through hands-on learning.
SDG 9	• **SDG 9: Industry, Innovation, and Infrastructure** This project develops a hardware-based cybersecurity solution. • Uses STM32 technology for real-time USB threat detection. • Supports safer and more reliable digital infrastructure.
SDG 16	• **SDG 16 – Peace, Justice and Strong Institutions** Protects systems from malicious USB attacks. • Promotes secure and responsible use of technology. • IHelps create a safer digital environment.

ABSTRACT

The proliferation of smartphones, smartwatches, and wireless earphones has created significant challenges in maintaining academic integrity within examination environments. This project proposes a low-cost, standalone Exam Hall Smartphone and Bluetooth Detector designed to identify the presence of active wireless devices within a restricted proximity. The system is developed using an ESP32 microcontroller, which integrates Wi-Fi and Bluetooth Low Energy (BLE) capabilities for passive wireless signal detection. The ESP32 continuously scans for 802.11 wireless probe requests and BLE advertising packets emitted by nearby devices. The received signal strength indicator (RSSI) is evaluated against a predefined threshold to determine whether a wireless device is likely to be within the monitored area.

When a detected wireless signal exceeds the configured RSSI threshold, the system provides an immediate visual indication through a red LED alarm and displays the corresponding status on a 16×2 I2C LCD. The hardware is implemented as a compact standalone system using an ESP32 development board, PCF8574-based LCD module, MB102 breadboard power supply, LED, and current-limiting resistor. A non-blocking firmware approach based on the millis() timer enables continuous scanning and simultaneous status indication without unnecessarily interrupting the detection process. The proposed system does not attempt to access, decrypt, or monitor the contents of private communications; it focuses only on detecting observable wireless activity and its signal strength. This provides a practical and privacy-conscious approach to assisting invigilators in maintaining examination integrity while demonstrating the application of embedded systems, wireless communication, cybersecurity, and real-time monitoring technologies.

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LIST OF ABBREVIATIONS

AR	: *AugmentedReality*
VR	: *VirtualReality*
eTAM	: *ExtendedTechnologyAcceptanceModel*

CHAPTER 1
INTRODUCTION

1.1 Overview

The rapid growth of smartphones, smartwatches, wireless earphones, and other portable wireless devices has created new challenges in maintaining academic integrity during examinations. Traditional examination invigilation mainly depends on physical observation by invigilators, which may not always be sufficient to identify electronic devices that are concealed or kept in silent mode. The use of wireless communication technologies makes it possible for unauthorized devices to remain connected or periodically transmit signals without attracting visual attention. Therefore, a technological solution capable of assisting invigilators in identifying nearby wireless device activity can improve the security and reliability of examination environments.

The Exam Hall Smartphone and Bluetooth Detector is a low-cost, standalone system designed to detect the presence of active wireless devices within a monitored examination area. The proposed system uses an ESP32 development board as the central processing and wireless detection unit. The ESP32 provides integrated Wi-Fi and Bluetooth Low Energy (BLE) capabilities, allowing the system to monitor observable wireless activity without requiring a separate sensor controller. The system is designed around a simple embedded architecture consisting of an ESP32, a 16×2 I2C LCD display, an LED indicator, and a regulated power supply.

The detector operates by passively scanning for wireless signals that are observable in its surroundings. On the Wi-Fi side, the system can identify wireless packet activity such as probe requests, while BLE scanning can identify advertising activity from compatible nearby devices. The system obtains the Received Signal Strength Indicator (RSSI) associated with detected

wireless activity. A signal whose RSSI exceeds a predefined threshold is treated as an indication of potentially nearby wireless device activity. The RSSI value is used as an approximate indicator of signal proximity rather than as an exact measurement of physical distance.

When the detected signal satisfies the configured detection condition, the system provides an immediate visual indication through a red LED and displays the corresponding status on the 16×2 I2C LCD. During normal operation, the LED can provide a periodic scanning indication while the LCD presents the current system state. When a stronger signal is detected, the system changes its alert state so that the invigilator can be notified without requiring continuous observation of the detector.

The current implementation follows a V2 ESP32 Standalone architecture. An earlier design considered a dual-microcontroller arrangement in which an STM32 Blue Pill acted as the main controller and an ESP32 was used for wireless sensing, with communication between the two devices through UART. The architecture was subsequently simplified by assigning both wireless detection and peripheral control to the ESP32. This reduces wiring complexity, eliminates the requirement for UART communication between two microcontrollers, and provides a more compact and stable implementation.

The proposed system is intended as an assistive detection mechanism rather than a replacement for human invigilation. It does not attempt to read messages, listen to conversations, access stored information, or decrypt private wireless communications. Its operation is limited to observing wireless activity and evaluating signal strength. This makes the project a practical application of embedded systems, wireless communication, cybersecurity, and real-time monitoring for supporting academic integrity in examination environments.

1.2 Objective

The main objective of this project is to design and implement a low-cost, standalone system for detecting observable Wi-Fi and Bluetooth/BLE wireless activity associated with nearby electronic devices in an examination environment. The system uses an ESP32-based wireless detection mechanism to passively scan for observable wireless activity and evaluate the received signal strength using RSSI-based threshold detection. It provides real-time detection information through a 16×2 I2C LCD and gives a visual alert through a red LED when the configured detection condition is met. The project aims to develop a compact and practical prototype that assists invigilators in maintaining academic integrity and examination security while demonstrating the application of ESP32, wireless communication, embedded programming, and cybersecurity principles.

1.3 Organisation of the Report

The report is organised as follows :

- **Chapter 1 : Introduction** discusses the background of wireless devices in examination environments, the proposed detection system, its objectives, and the organisation of the report.
- **Chapter 2 : Literature Review**discusses existing approaches related to Wi-Fi, Bluetooth/BLE, RSSI-based detection, and wireless monitoring.
- **Chapter 3: Design and Implementation** describes the system design, working principle, architecture, and implementation.
- **Chapter 4 :Hardware and Software Specification**presents the hardware components and software requirements used in the project.
- **Chapter 5 : Results and Discussion**presents the experimental results and discusses the

performance of the developed system.

- **Chapter 6 : Conclusion and Future Scope** summarises the project and discusses possible future improvements.
- **Chapter 7: Code** contains the program code used to implement the system.
- **Chapter 8: References** lists the sources and references used for the project.
- **References** lists out references for Augmented Reality and Virtual Reality for learning.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Wireless communication technologies such as Wi-Fi and Bluetooth are widely used in smartphones, laptops, smartwatches, and other electronic devices. The increasing use of wireless-enabled devices has created a need for systems that can observe and monitor wireless activity in different environments. In examination halls, monitoring wireless activity can provide additional information to invigilators when unauthorized wireless devices are present.

Traditional examination monitoring mainly depends on direct observation by invigilators. Although manual observation is useful, it may be difficult to continuously identify wireless activity from devices that are not easily visible. Wireless detection systems provide an additional approach by scanning the surrounding environment and identifying observable Wi-Fi and Bluetooth/BLE signals.

The proposed project combines Wi-Fi detection and Bluetooth/BLE detection with an ESP32 microcontroller to provide a compact wireless activity monitoring system. The ESP32 performs wireless scanning and processes information such as received signal strength. A 16×2 I2C LCD is used to display the system status, while a red LED provides a visual indication when the configured detection condition is satisfied.

2.2 Wi-Fi Detection

Wi-Fi is one of the most commonly used wireless communication technologies in smartphones, laptops, and other electronic devices. Wi-Fi scanning can be used to identify observable wireless networks in the surrounding environment. Information such as network name and

received signal strength can be obtained during the scanning process.

In a wireless monitoring system, the ESP32 can perform Wi-Fi scanning to observe nearby wireless activity. The detected information can be processed according to predefined conditions. RSSI values obtained during scanning provide information about the relative strength of the detected wireless signal.

The proposed system uses the Wi-Fi scanning capability of the ESP32 to monitor observable wireless activity in the examination environment. When the detected information satisfies the configured detection condition, the system provides an indication through the LCD and red LED.

2.3 Bluetooth and BLE Detection

Bluetooth is another widely used wireless communication technology found in smartphones, wireless accessories, smartwatches, and other electronic devices. Bluetooth Low Energy (BLE) is designed for low-power wireless communication and is also commonly supported by modern devices.

Bluetooth and BLE scanning can be used to observe nearby compatible wireless devices. During the scanning process, the system can obtain information related to detected devices and their signal strength. This information can then be processed by the microcontroller according to the required detection conditions.

The proposed system uses the Bluetooth/BLE capability of the ESP32 to monitor nearby wireless activity. The detected Bluetooth/BLE information is processed by the ESP32, and the system provides an alert when the configured detection condition is satisfied.

2.4 RSSI-Based Wireless Detection

Received Signal Strength Indicator (RSSI) is a parameter that represents the received strength of a wireless signal. RSSI can be used to obtain an approximate indication of how strong a detected wireless signal is at the location of the monitoring device.

RSSI-based detection can be implemented by comparing the measured signal strength with a predefined threshold. When the measured RSSI satisfies the configured threshold condition, the system can identify the signal as requiring further attention. The RSSI value can also change depending on distance, obstacles, interference, and the surrounding environment.

In the proposed system, RSSI information obtained from Wi-Fi and Bluetooth/BLE scanning is processed by the ESP32. A predefined RSSI threshold is used as part of the detection logic. When the detected signal satisfies the specified condition, the LCD displays the corresponding status and the red LED is activated.

2.5 ESP32-Based Wireless Monitoring

Microcontrollers are widely used in embedded systems because they can acquire information, process inputs, and control output devices. A microcontroller with integrated wireless communication provides a compact platform for developing wireless monitoring applications.

The ESP32 is used as the main processing unit in the proposed project. It provides Wi-Fi and Bluetooth/BLE capabilities along with general-purpose input and output interfaces. The ESP32 receives wireless scanning information, processes the detected signals, and controls the output devices according to the programmed detection logic.

The use of an ESP32-based architecture allows wireless detection, signal processing, status display, and alert functions to be integrated into a single compact embedded system. This reduces the need for separate wireless modules and provides a simple platform for implementing the

proposed detector.

2.6 Display and Alert System

An effective monitoring system should provide a clear indication when a configured wireless detection condition is identified. Visual output devices can help the invigilator observe the operating condition of the system without continuously checking the serial monitoring interface.

The proposed system uses a 16×2 I2C LCD for displaying information about the current operating condition. The LCD can indicate scanning status and provide messages when wireless activity is detected. The I2C interface allows the display to communicate with the ESP32 using a small number of connections.

A red LED is used as a visual alert indicator. When the programmed detection condition is satisfied, the ESP32 activates the LED to indicate detected wireless activity. The combination of the LCD and LED provides a simple and easily observable alert mechanism for the examination environment.

2.7 Existing Approach and Proposed Approach

Conventional examination monitoring mainly depends on manual observation by invigilators. This approach requires continuous attention and may not provide direct information about wireless activity from nearby electronic devices.

The proposed system introduces an embedded wireless monitoring approach by combining Wi-Fi scanning, Bluetooth/BLE scanning, RSSI-based processing, and visual alert mechanisms. The main differences between the conventional and proposed approaches are summarized below.

Conventional Monitoring	Proposed System
Primarily depends on manual observation	Uses wireless scanning for continuous monitoring
Wireless activity may be difficult to identify visually	ESP32 scans observable Wi-Fi and Bluetooth/BLE activity
No direct RSSI-based information	RSSI is used as part of the detection condition
Alerts depend mainly on invigilator observation	Red LED provides a visual alert
Limited automated wireless processing	ESP32 processes detected wireless information
System status may not be displayed continuously	16×2 I2C LCD provides system information

Figure 2.1: Comparison of Conventional and Proposed wireless Monitoring Approches

2.8 Review Summary

The reviewed concepts indicate that Wi-Fi detection, Bluetooth/BLE detection, RSSI-based processing, microcontroller-based wireless monitoring, and alert mechanisms can be combined to support examination-hall wireless activity monitoring.

The proposed project brings these functions together using an ESP32 microcontroller, Wi-Fi and Bluetooth/BLE scanning, RSSI-based detection, a 16×2 I2C LCD, and a red LED. The system is intended to observe nearby wireless activity and provide an indication when the configured detection condition is satisfied.

The literature review therefore provides the foundation for the design and implementation of the proposed Exam Hall Smartphone and Bluetooth Detector. The ESP32-based architecture provides a compact approach for integrating wireless detection, processing, display, and alert functions within a single embedded system.

CHAPTER 3
DESIGN AND IMPLEMENTATION

The proposed system is designed to detect observable Wi-Fi and Bluetooth/BLE wireless activity using an ESP32 microcontroller. The system is intended to assist in monitoring examination environments by providing an indication when the configured wireless detection condition is satisfied. The ESP32 performs wireless scanning and processes the received signal information using an RSSI-based threshold.

The system consists of an ESP32, Wi-Fi and Bluetooth/BLE scanning functions, a 16×2 I2C LCD, and a red LED indicator. The ESP32 acts as the main processing unit and continuously scans for observable wireless activity. The detected signal information is processed and the system status is displayed on the LCD, while the red LED provides a visual alert when the required condition is detected.

The implementation uses a non-blocking millis()-based timing approach to allow the scanning and indication functions to operate efficiently. The system is designed as a compact and low-cost standalone prototype, providing real-time wireless activity information without accessing private communication content.

3.1 System Design

The system design focuses on wireless activity detection, signal processing, and real-time indication. The ESP32 scans for Wi-Fi and Bluetooth/BLE activity and obtains available signal information such as RSSI. The RSSI value is compared with the configured threshold, and the corresponding detection status is provided through the LCD and LED.

3.2 System Architecture

The different classifications is as follows:

3.2.1 Classification by Detection Type

1. **Wi-Fi Detection:** For Wi-Fi detection purposes, the ESP32 scans for observable wireless activity associated with Wi-Fi communication. The detected information includes available signal details such as RSSI, which is used for further processing.
2. **Bluetooth/BLE Detection:** For Bluetooth/BLE detection purposes, the ESP32 observes nearby BLE wireless activity from compatible electronic devices. The available signal information is collected and processed according to the configured detection condition.
3. **Alert and Monitoring:** For alert and monitoring purposes, the primary goal is to provide immediate information to the invigilator when the configured wireless detection condition is satisfied. The 16×2 I2C LCD displays the system status, while the red LED provides a visual indication of detected wireless activity.

3.2.2 Classification by System Response

1. **Monitoring:** The system continuously monitors nearby wireless activity and collects relevant signal information for further processing.
2. **Detection:** The detected Wi-Fi and Bluetooth/BLE signals are analyzed according to the configured detection conditions to identify possible unauthorized wireless activity.
3. **Alert Generation:** When a configured condition is satisfied, the system provides an immediate alert through the 16×2 I2C LCD and red LED, helping the invigilator monitor the examination environment.

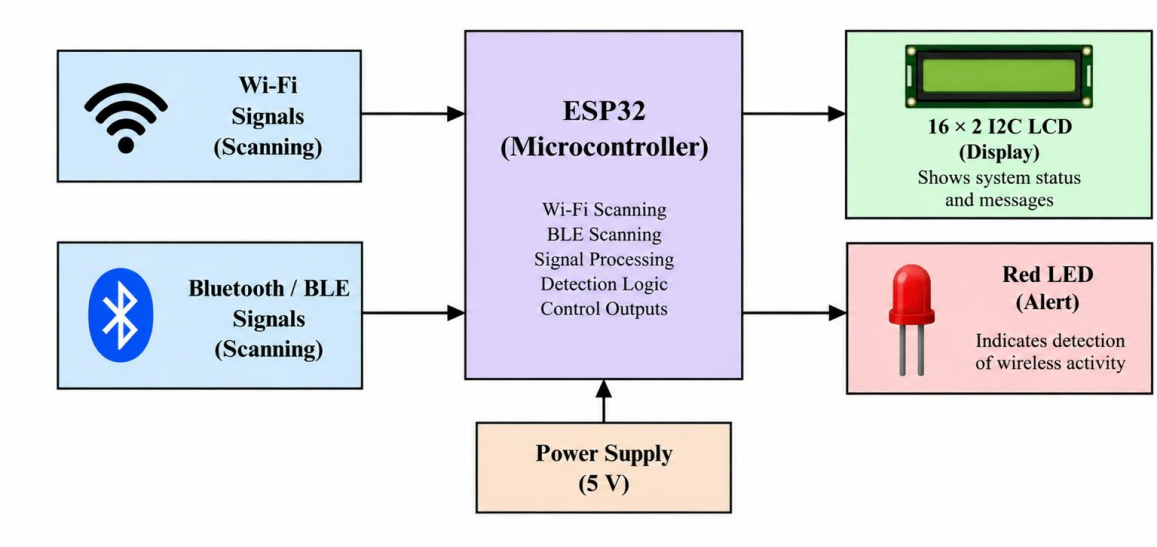


Figure 3.1: Block Diagram Of Exam Hall Smartphone and Bluetooth Detector

3.2.3 Classification by Implementation Technology

1. **Embedded System Technology:** The system is implemented using an ESP32 microcontroller, which acts as the main processing unit. It performs wireless scanning, collects signal information, and processes the detected Wi-Fi and Bluetooth/BLE activity according to the programmed conditions.

2. **Wireless Detection Technology:** The ESP32 uses its built-in wireless capabilities to observe nearby Wi-Fi and Bluetooth/BLE activity. The detected wireless information, including RSSI values, is used to identify and analyze suspicious or unauthorized wireless activity within the examination environment.

3. **Alert and Display Technology:** The system uses a 16×2 I2C LCD and a red LED to provide real-time status and alert indications. The LCD displays the current system condition, while the LED provides a visual indication when the configured wireless detection condition is detected.

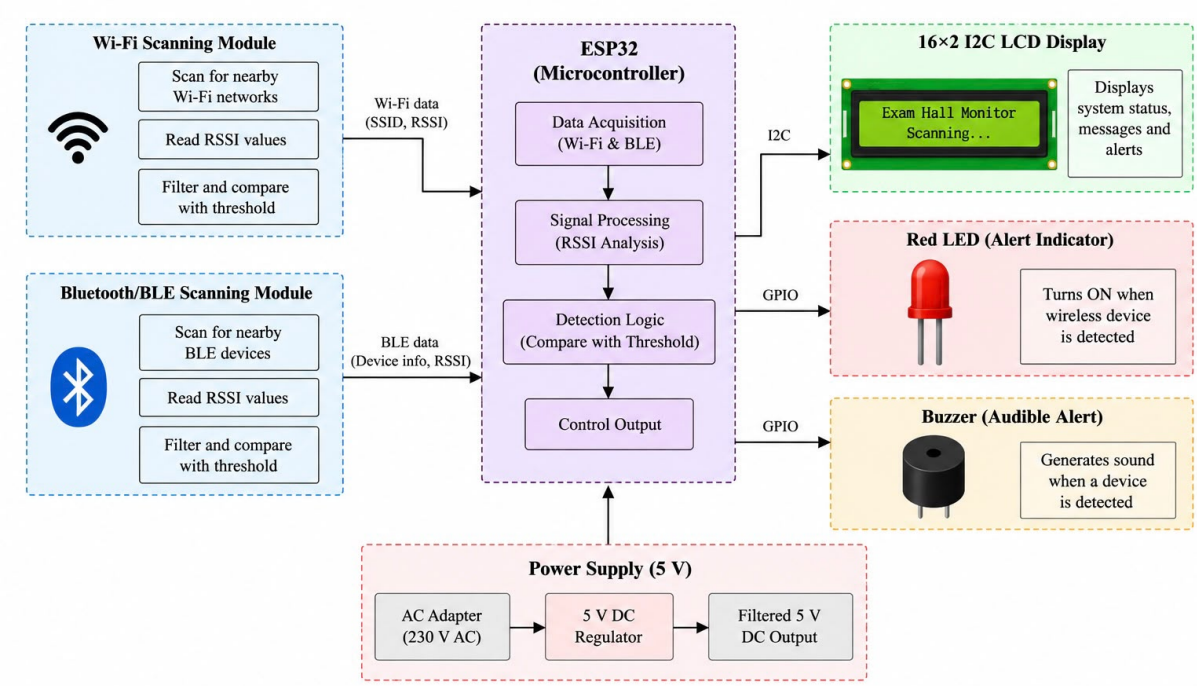


Figure 3.2: Detailed Block Diagram Of Exam Hall Smartphone and Bluetooth Detector

3.3 Working Principle

3.3.1 Wi-Fi Detection

The ESP32 continuously scans the surrounding environment for observable Wi-Fi activity. It collects available wireless information, including the received signal strength (RSSI), and processes the detected signals according to the configured conditions. The processed information is used to identify wireless activity that may require attention during the examination.

3.3.2 Bluetooth/BLE Detection

The ESP32 also monitors nearby Bluetooth/BLE wireless activity from compatible electronic devices. The detected signal information is collected and analyzed along with the Wi-Fi data. This allows the system to monitor multiple types of wireless communication within the examination environment.

3.3.3 Alert and Monitoring

When the configured detection condition is satisfied, the system provides an immediate alert to the invigilator. The 16×2 I2C LCD displays the current system status, while the red LED provides a visual indication of detected wireless activity. The system continuously monitors the environment and assists the invigilator in identifying unauthorized wireless activity.

3.4 Architecture

The architecture of the USB Attack Detection and Auto Isolation System consists of a microcontroller-based detection unit, wireless monitoring components, an alert and display unit, and the connected system interface. The ESP32 acts as the main processing unit of the system. It is responsible for scanning and processing nearby Wi-Fi and Bluetooth/BLE wireless activity and evaluating the detected information according to the configured detection conditions.

The wireless detection unit collects available signal information such as RSSI from nearby devices. The collected information is processed by the ESP32 to identify wireless activity that satisfies the specified detection criteria. The system is designed to continuously monitor the surrounding environment without requiring manual operation during the detection process.

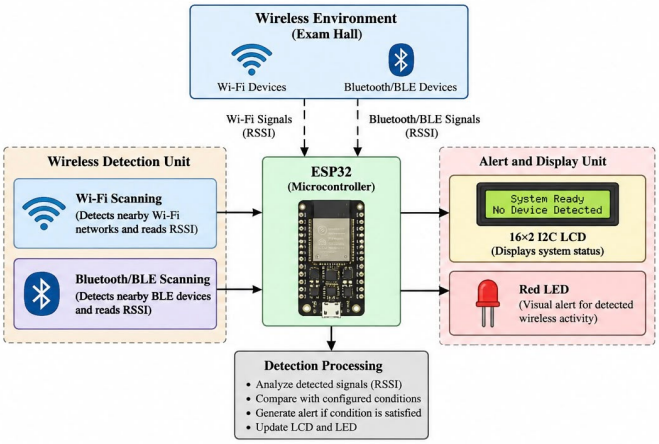


Figure 3.3: Architecture of the Wi-Fi and Bluetooth/BLE Detection and Alert System

The alert and monitoring section consists of a 16×2 I2C LCD and a red LED. The LCD provides information about the current operating condition of the system, while the LED provides a visual indication when the configured wireless detection condition is detected. The combination of these components enables the invigilator to observe the system status easily during operation.

The overall system architecture connects the detection, processing, and alert units in a continuous sequence. The ESP32 receives wireless information, processes the detected signals, and activates the corresponding output when the specified condition is satisfied. This architecture provides a simple and compact approach for monitoring unauthorized wireless activity in an examination environment.

CHAPTER 4
HARDWARE AND SOFTWARE SPECIFICATION

4.1 Hardware Requirements

- ESP32 Development Board :acts as the main processing and control unit of the system. It provides built-in Wi-Fi and Bluetooth/BLE capabilities, which are used to detect nearby wireless activity. The ESP32 processes the detected signal information and evaluates it according to the configured detection conditions.
- 16×2 I2C LCD Display:is used to provide information about the current operating condition of the system. It can display system status and detection-related messages to the invigilator. The I2C interface reduces the number of connections required between the LCD and ESP32.
- Red LED:is used as a visual alert indicator. When the configured wireless detection condition is satisfied, the ESP32 activates the LED to provide an immediate visual indication to the invigilator.
- Resistor for LED:A resistor is connected in series with the red LED to limit the current flowing through the LED and protect it from excessive current.
- Connecting Wires:Connecting wires are used to establish electrical connections between the ESP32, LCD, LED, resistor, and other circuit components. They provide the required signal and power connections during system implementation.
- Breadboard:A breadboard is used for assembling and testing the circuit without permanent soldering. It provides a convenient platform for connecting the ESP32 with the LCD and alert components during development.

- USB Cable and Power Supply:A USB cable is used for programming the ESP32 and supplying power to the development board. The power supply enables continuous operation of the system during testing and deployment.

4.2 Software Requirements

- Arduino IDE is used for writing, compiling, and uploading the program to the ESP32 Development Board. It provides an easy environment for developing and testing the embedded program.
- The ESP32 Board Package provides the necessary support for programming the ESP32 using the Arduino IDE. It enables access to the hardware functions required for wireless scanning, input/output control, and other ESP32 features.
- The Wi-Fi and Bluetooth/BLE libraries are used to implement wireless scanning and detection. These libraries allow the ESP32 to obtain available wireless signal information from nearby devices and process it according to the programmed conditions. LiquidCrystal I2C library is used to interface the 16×2 I2C LCD with the ESP32. It provides functions for displaying system status and detection messages on the LCD.
- Embedded C/C++ is used to develop the main control program of the system. The program performs wireless scanning, processes the detected information, evaluates the configured conditions, updates the LCD, and controls the red LED when a detection condition is satisfied.

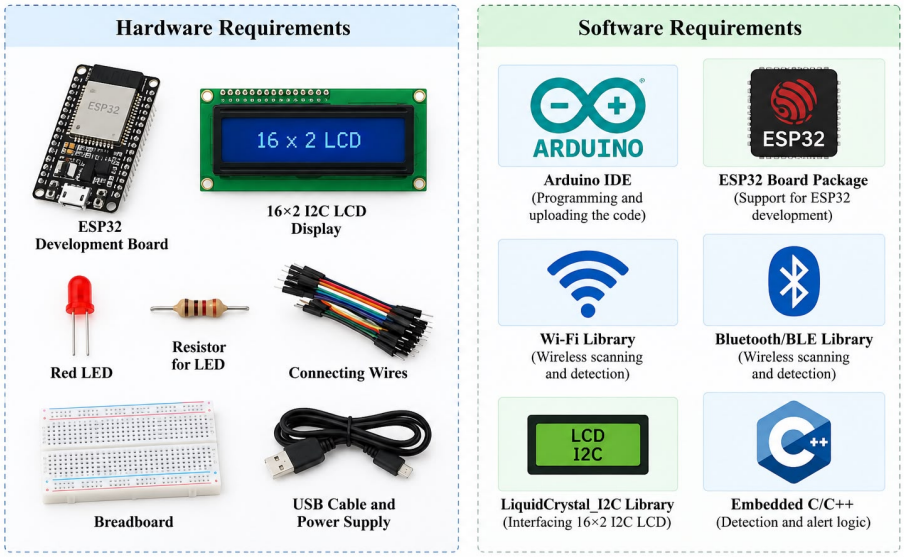


Figure 4.1: Hardware and Software Requirements for the Wi-Fi and Bluetooth/BLE Detection and Alert System

CHAPTER 5
RESULTS AND DISCUSSION

The implemented Wi-Fi and Bluetooth/BLE Detection and Alert System was designed to monitor wireless activity in an examination environment. The ESP32 performs wireless scanning and processes the detected signal information according to the configured detection conditions. The system provides the detection status through the 16×2 I2C LCD and gives a visual alert using the red LED.

5.1 Experimental Setup

The developed system was assembled using the ESP32 Development Board, 16×2 I2C LCD, red LED, resistor, breadboard, connecting wires, and USB power supply. The ESP32 was programmed using the Arduino IDE with the required Wi-Fi and Bluetooth/BLE libraries.

During testing, the system was powered through the USB connection and placed in an environment containing nearby wireless devices. The ESP32 continuously scanned the surrounding wireless environment and processed the available signal information. The LCD was used to observe the system status, while the red LED indicated the configured detection condition.

5.2 Wi-Fi Detection Results

The Wi-Fi detection function was tested by scanning for nearby Wi-Fi activity using the ESP32. The system was able to identify observable Wi-Fi signals and obtain available information such as signal strength (RSSI). The detected wireless information was processed by the ESP32 according to the programmed conditions. When the specified condition was satisfied, the system updated the LCD and activated the alert indicator. This demonstrates the ability of the ESP32 to monitor nearby Wi-Fi activity continuously.

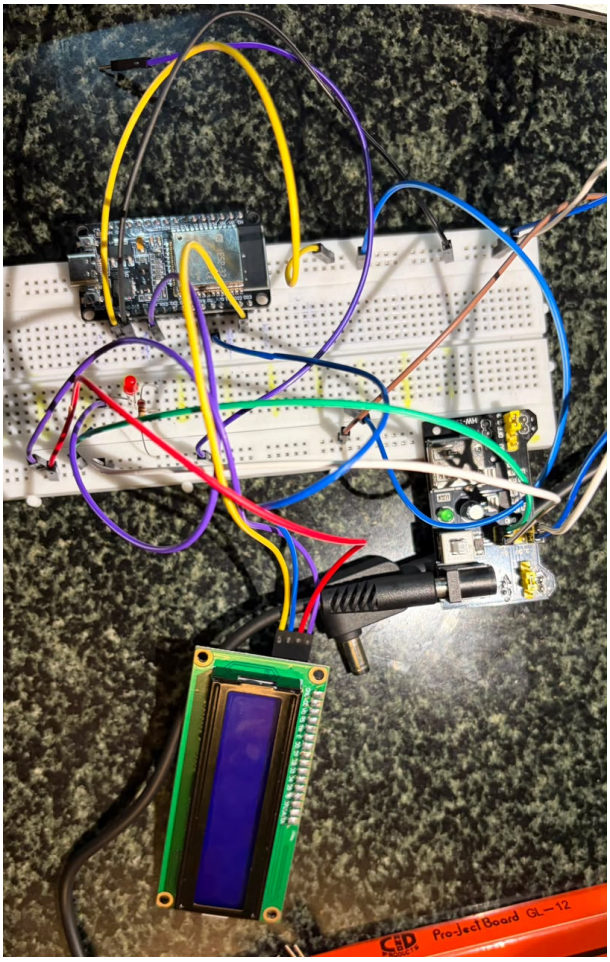


Figure 5.1: Experimental Setup

5.3 Bluetooth/BLE Detection Results

The Bluetooth/BLE detection function was tested using nearby Bluetooth/BLE-enabled devices. The ESP32 scanned the surrounding environment and processed the detected wireless activity.

The system was able to distinguish Bluetooth/BLE activity from the available wireless signals and process the detected information according to the configured detection conditions. The detection status was provided through the LCD and the alert mechanism.

5.4 RSSI-Based Detection Results

RSSI (Received Signal Strength Indicator) was used as one of the available signal parameters for evaluating detected wireless activity. The RSSI value provides an indication of the relative strength of a detected wireless signal.

The ESP32 obtains the available RSSI information during wireless scanning and uses it as part of the programmed detection logic. Changes in signal strength can help in identifying wireless devices operating within the surrounding examination environment.

5.5 LCD and LED Alert Results

The 16×2 I2C LCD successfully provides the operating status and detection information of the system. It allows the invigilator to monitor the system without requiring interaction with the ESP32.

The red LED acts as a visual alert when the configured wireless detection condition is satisfied. The combination of the LCD and LED provides both status information and an immediate visual indication of detected wireless activity.

5.6 Discussion

The developed system provides a compact approach for monitoring Wi-Fi and Bluetooth/BLE activity in an examination environment. The ESP32 combines wireless scanning, signal processing, and alert control in a single development board, reducing the need for additional wireless detection hardware. The use of RSSI provides additional signal information that can be considered during detection. The LCD makes the system status easy to observe, while the LED provides a quick visual alert when the configured condition is detected.

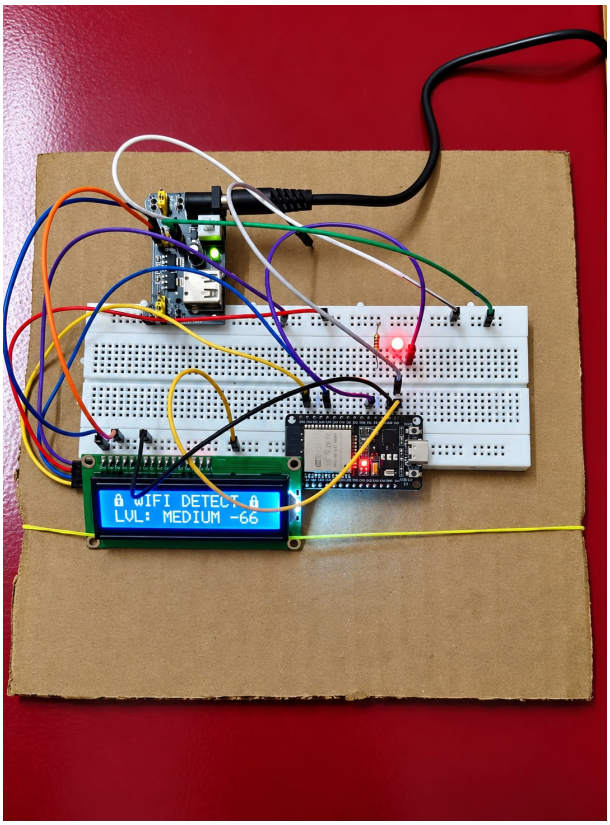


Figure 5.2: Wi-Fi Detection with RSSI Level and LED Alert

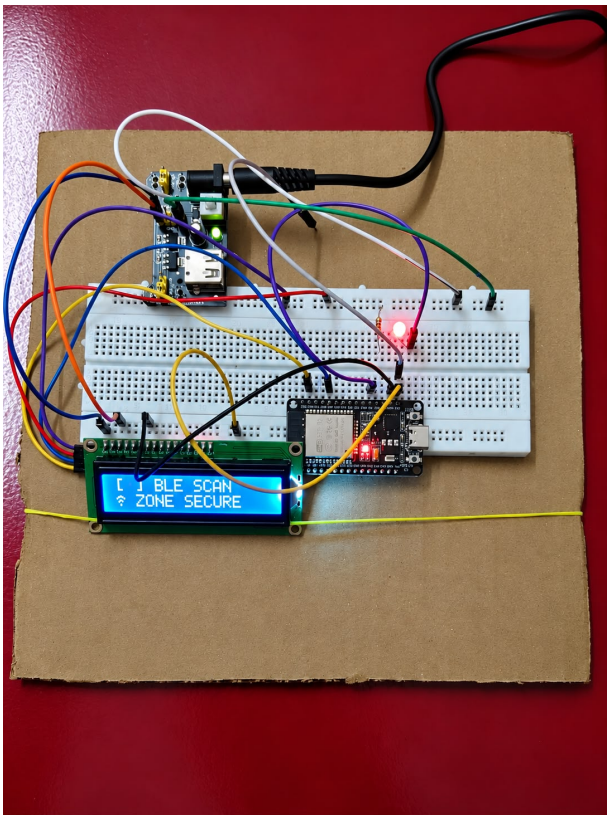


Figure 5.3: Bluetooth/BLE Scanning and Secure-Zone Status Display

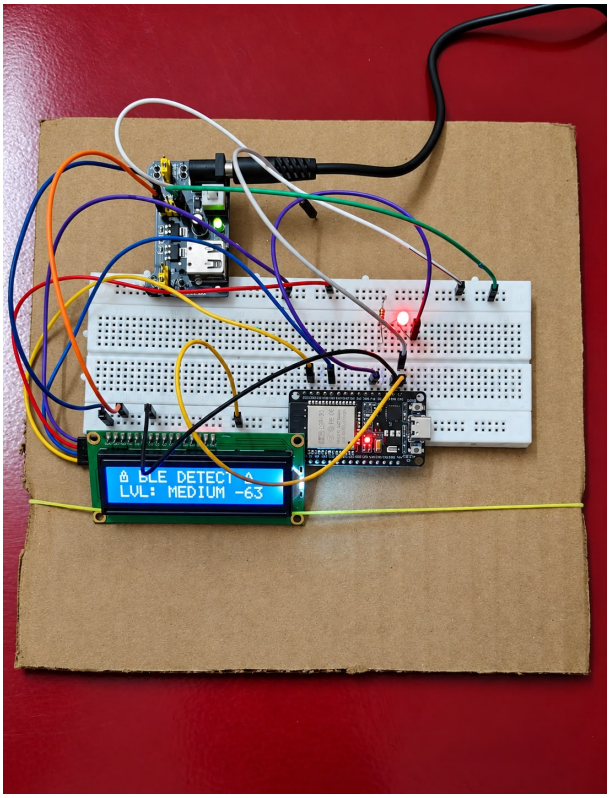


Figure 5.4: Bluetooth/BLE Detection with RSSI Level and LED Alert

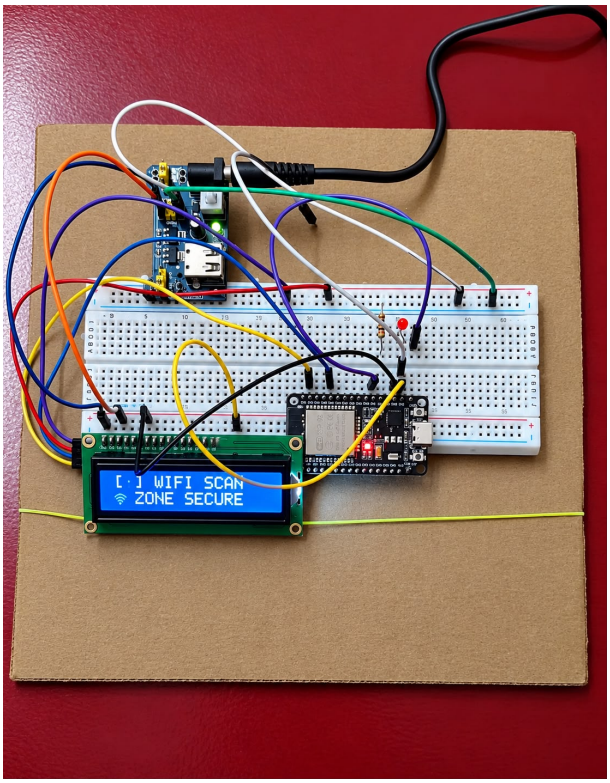


Figure 5.5: Wi-Fi Scanning and Secure-Zone Status Display

Test	Observation	Result
Wi-Fi Detection	Nearby Wi-Fi activity detected.	Successfully detected
Bluetooth/BLE Detection	Nearby BLE activity detected.	Successfully detected
RSSI Analysis	Signal strength (RSSI) obtained and analyzed.	RSSI information obtained successfully
LCD Display	System status and detection information displayed on LCD.	Displayed successfully
Red LED	Visual alert provided when a device is detected.	LED turns ON successfully
Continuous Monitoring	Continuous wireless monitoring demonstrated.	Working as expected

Figure 5.6: Test Results and Observations

S. No.	Detection Type	RSSI Value (dBm)	Signal Strength Level	Observation
1	**Wi-Fi Detection**	**-66**	**Medium**	Wi-Fi signal detected
2	**Bluetooth/BLE Detection**	**-63**	**Medium**	BLE signal detected
3	**Wi-Fi Scan**	Not detected	—	No signal detected above the configured threshold
4	**Bluetooth/BLE Scan**	Not detected	—	No signal detected above the configured threshold

Figure 5.7: RSSI Output Results

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Wi-Fi and Bluetooth/BLE Detection and Alert System was developed to monitor wireless activity in an examination environment using the ESP32 Development Board. The ESP32 acts as the main processing unit and performs Wi-Fi and Bluetooth/BLE scanning to detect nearby wireless activity.

The detected wireless information, including available signal strength (RSSI), is processed according to the configured detection conditions. The 16×2 I2C LCD provides the current system status and detection information, while the red LED provides a visual alert when the specified wireless detection condition is satisfied.

The implemented system provides a simple, compact, and low-cost approach for assisting invigilators in monitoring wireless activity during examinations. The testing demonstrates that the ESP32 can perform wireless scanning and provide immediate status and alert indications through the connected output devices.

However, the effectiveness of wireless detection can be affected by factors such as signal strength, distance from the device, surrounding wireless interference, and the type of wireless device being detected. Therefore, the system should be considered as a monitoring and alert mechanism for assisting examination supervision.

6.2 Future Scope

The proposed Wi-Fi and Bluetooth/BLE Detection and Alert System can be further improved by incorporating additional detection and monitoring capabilities. The wireless detec-

tion mechanism can be enhanced to improve the identification of different Wi-Fi and Bluetooth/BLE devices and provide more reliable monitoring in examination environments.

The signal analysis can also be improved by using more detailed RSSI and other available wireless parameters. This can help in analysing detected wireless activity more effectively and provide additional information about nearby devices.

Data logging can be incorporated in future versions to store detected wireless activity along with time and signal information. Such records can be useful for later analysis and maintaining examination-related monitoring information.

The existing 16×2 I2C LCD can be replaced with a larger or more advanced display to provide detailed information about detected wireless activity. Additional alert mechanisms, such as a buzzer or remote notification, can also be added along with the existing red LED to provide immediate alerts.

The system can also be extended to support network-based monitoring, where multiple detection units can be connected to a central monitoring system. This would allow wireless activity from different examination areas to be monitored from a single location.

Finally, the system can be developed into a more compact and portable unit with an integrated power source. Such improvements can make the system easier to deploy and support continuous wireless monitoring while maintaining a simple and practical design.

CHAPTER 7
CODE

1. WI-FI DETECTION ROUTINE

```
if (currentMode == WIFI_MODE) {
  int scanStatus = WiFi.scanComplete();

  // Background Wi-Fi scan completed successfully
  if (scanStatus >= 0) {
    int bestRSSI = -100;

    // Evaluate peak signal strength among detected access points
    for (int i = 0; i < scanStatus; i++) {
      if (WiFi.RSSI(i) > bestRSSI) {
        bestRSSI = WiFi.RSSI(i);
      }
    }
    WiFi.scanDelete(); // Free memory buffer

    // Switch state to BLE mode to share the 2.4 GHz radio
    currentMode = BLE_MODE;
    bleScanStarted = false;
    bleScanDone = false;

    // Check against sensitivity threshold (-85 dBm)
    if (bestRSSI >= WIFI_ALERT_THRESHOLD) {
      triggerAlert("WIFI", bestRSSI);
    }
  }
  else if (scanStatus == WIFI_SCAN_FAILED) {
    WiFi.scanNetworks(true, true); // Re-trigger async scan on failure
  }
}
```

2. BLUETOOTH LOW ENERGY (BLE) DETECTION ROUTINE

```
else if (currentMode == BLE_MODE) {

  // Initiate a 1-second non-blocking BLE scan
  if (!bleScanStarted) {
    pBLEScan->clearResults();
    pBLEScan->start(1, bleScanCompleteCallback, false);
```

```
    bleScanStarted = true;
    bleScanDone = false;
  }
  // Process results once the asynchronous callback fires
  else if (bleScanDone) {
    int bestRSSI = -100;
    BLEScanResults* foundDevices = pBLEScan->getResults();

    // Parse RSSI for all discovered BLE peripherals (phones, smartwatches,
    earbuds)
    for (int i = 0; i < foundDevices->getCount(); i++) {
      int rssi = foundDevices->getDevice(i).getRSSI();
      if (rssi > bestRSSI) {
        bestRSSI = rssi;
      }
    }
    pBLEScan->clearResults(); // Prevent ESP32 heap memory exhaustion

    // Switch state back to Wi-Fi mode
    currentMode = WIFI_MODE;

    if (bestRSSI >= BLE_ALERT_THRESHOLD) {
      triggerAlert("BLE", bestRSSI);
    } else {
      // Area clear; immediately re-arm Wi-Fi scan
      WiFi.scanNetworks(true, true);
    }
  }
}
```

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